SVR COLLEGE OF COMMERCE AND MANAGEMENT STUDIES

Affiliated by Bangalore University & Approved by AICTE & MHRD , New Delhi

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A PROJECT SYNOPSIS ON

"SMART ATTENDANCE SYSTEM USING FACE RECOGNITION"

Submitted in partial fulfilment of the requirements for the Award of the Degree of VI Semester BCA, Bangalore University

Project Synopsis by :

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2025 – 2026

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1. Abstract / Executive Summary

The *Smart Attendance System Using Face Recognition* is an automated and intelligent system designed to replace traditional attendance marking methods in educational institutions and organizations.
The system uses computer vision and machine learning techniques to identify individuals based on facial features and automatically records their attendance in a centralized database.

The proposed system captures live images through a camera, detects faces, performs face recognition and updates attendance records in real time. This approach significantly reduces manual effort, eliminates proxy attendance and improves the accuracy and reliability of attendance records.

The system also provides administrative control, report generation and data analysis facilities for effective attendance management.

2. Introduction

Attendance management is an essential activity in educational institutions and corporate environments. Traditional attendance systems such as manual roll calls, signature sheets and card-based methods are inefficient, time consuming and prone to manipulation.

With recent advancements in artificial intelligence and computer vision, face recognition has emerged as a reliable biometric technology for personal identification. The proposed Smart Attendance System uses this technology to automatically recognize individuals and mark their attendance without physical interaction.

This system is suitable for:

- Schools and colleges
- Universities and training institutes
- Corporate offices
- Conferences and workshops

3. Problem Statement

The existing attendance systems suffer from several limitations such as:

- Manual attendance consumes significant time during classes and meetings
- Proxy attendance is easy in traditional systems
- Human errors occur during manual data entry
- Paper-based records are difficult to store and maintain
- Data retrieval and analysis become complex
- Card and ID based systems can be misused
- Scalability becomes difficult when the number of users increases

Hence, there is a strong requirement for a secure, automated and scalable attendance solution.

4. Objectives of the Project

Primary Objectives

- To design and develop an automated attendance system using face recognition
- To prevent proxy attendance through biometric verification
- To minimize manual effort and human errors
- To maintain a centralized attendance database

Secondary Objectives

- To provide real-time attendance monitoring
- To generate attendance reports and analytics
- To develop a user-friendly administrative interface
- To ensure scalability and reliability of the system
- To achieve high accuracy in face recognition

5. Scope of the Project

Included Features

- Real-time face detection and recognition
- Automated attendance recording
- Student / employee registration and management
- Administrator dashboard
- Attendance reports and data analytics
- Role-based access control

Future Enhancements

- Mobile application support
- Cloud based deployment
- Integration with institutional ERP systems
- Multi-camera support
- Advanced AI based analytics

6. Literature Review / Existing Systems

Manual Attendance Systems

These systems are based on roll calls or paper sheets. They are time consuming and prone to proxy attendance and human errors.

Card Based Attendance Systems

RFID cards and barcode systems reduce manual effort but cards can be lost, shared or stolen.

Biometric Systems

Fingerprint and iris based systems improve security but require physical contact and dedicated hardware.

Existing Face Recognition Systems

Commercial systems provide good accuracy but are costly and offer limited customization.

Research Gap

There is a need for a low-cost, customizable, contactless and easy-to-deploy face recognition attendance system suitable for academic environments.

7. Proposed System

The proposed system uses a camera and artificial intelligence algorithms to detect and recognize faces and automatically mark attendance.

Key Features

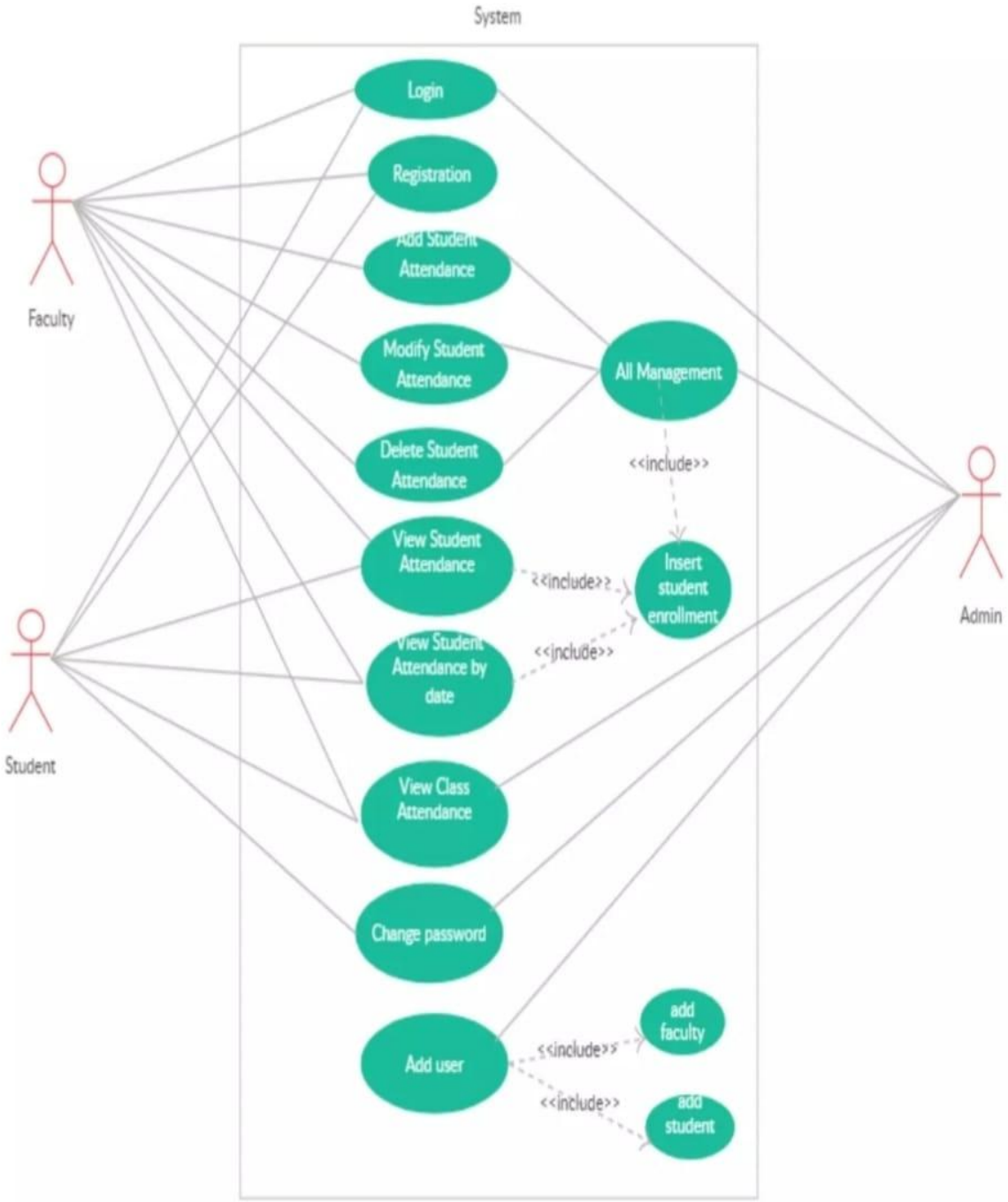
- Automated face detection
- Face recognition using trained models
- Real-time attendance recording
- User and profile management
- Secure data storage
- Reporting and analytics dashboard

Advantages

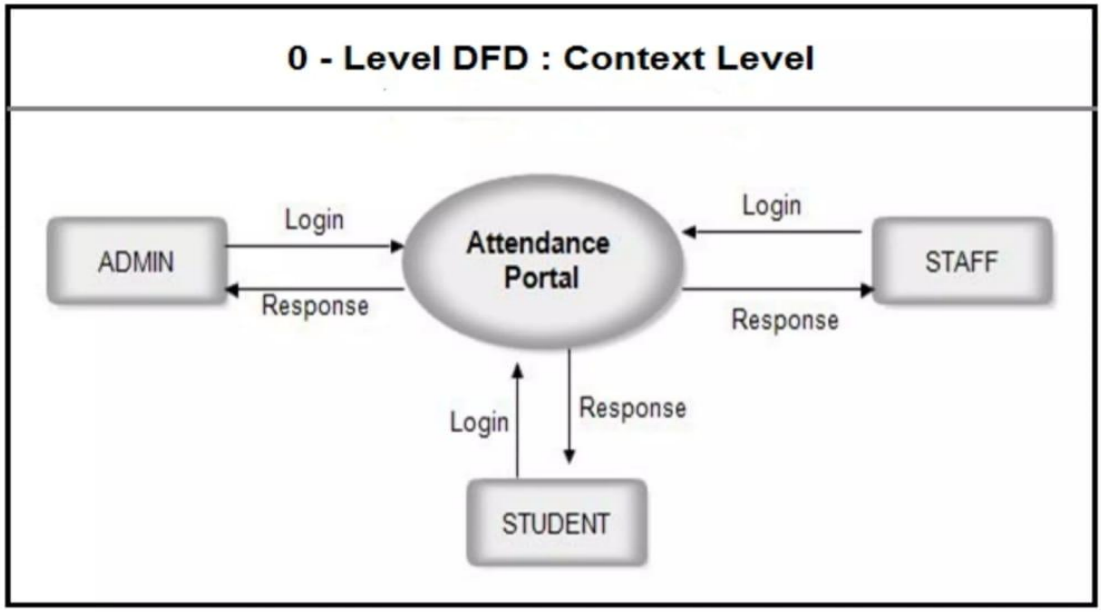
- Contactless operation
- Reduced operational cost
- High accuracy
- Real-time processing
- Scalable architecture
- Easy integration with existing systems

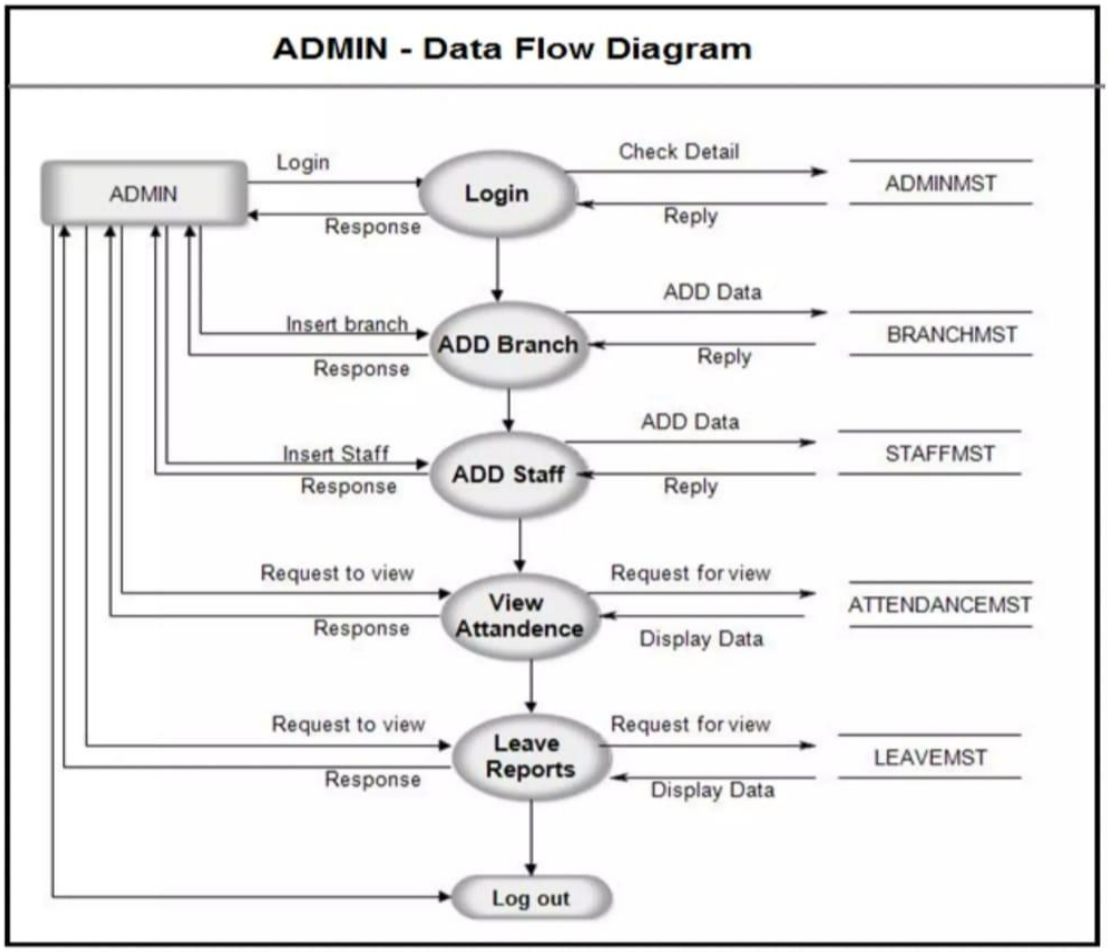
8. System Design

8.1 USE CASE DIAGRAM:



8.2: DFD-DIAGRAM:





09. Methodology / Working of the System

1. The system initializes the camera, database and recognition models
2. Live video frames are captured continuously
3. Faces are detected in the frames
4. Detected faces are pre-processed
5. Facial features are extracted
6. Extracted features are matched with stored face encodings
7. Attendance is marked automatically for recognized users
8. Records are stored with date and time information

10. Modules of the System

10.1 Registration Module

- New user enrollment
- Face sample collection
- Profile management

10.2 Face Detection and Recognition Module

- Real-time face detection
- Face identification
- Confidence threshold handling

10.3 Attendance Marking Module

- Automatic attendance entry
- Time and date stamping
- Duplicate entry prevention

10.4 Reporting Module

- Daily and monthly reports
- Custom date range reports
- Export to PDF and Excel

10.5 Admin Dashboard

- User management
- System configuration
- Data backup and maintenance
- Security and access control

11. Technology Stack

Programming Languages

- Python
- HTML, CSS, JavaScript

Frameworks and Libraries

- Flask
- OpenCV
- face_recognition
- Bootstrap
- Jinja2

Database

- MySQL

Development Tools

- Visual Studio Code
- Git & Github
- Postman
- Docker

Hardware Requirements

- Webcam
- Minimum Intel i3 / equivalent processor
- 4 GB RAM (8 GB recommended)

Software Requirements

- Python 3.8 or above
- MySQL Server
- Web browser

12. Implementation Plan

Phase 1 – Setup and Design

- Environment configuration
- Database design
- Initial UI design

Phase 2 – Core Development

- Face detection and recognition
- Registration module
- Attendance processing

Phase 3 – Integration and Optimization

- Reporting module
- Security features
- Performance optimization

Phase 4 – Testing and Deployment

- System testing
- Bug fixing
- Final deployment and documentation

13. Expected Outcomes

- Significant reduction in attendance processing time
- Elimination of proxy attendance
- Accurate and reliable attendance records
- Automated reporting facilities

14. Conclusion

The Smart Attendance System Using Face Recognition provides an efficient, secure and automated solution for attendance management. By eliminating manual procedures and introducing biometric verification through facial recognition, the system ensures high accuracy, improved operational efficiency and better data management.

The proposed system offers a practical and scalable solution suitable for academic institutions and organizations.